

When the Attack Is Not an Attack: Validity Failures in LLM-Generated Red-Teaming

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Abstract

Language models are increasingly used to generate the probes that test other models. The appeal is scale and fluency: an LLM can rewrite an adversarial example into text a person might actually write. This paper asks what such a pipeline is measuring. I identify three ways a generated probe can be counted in a red-teaming result without being a valid test, and I quantify each on 32 model–attack–dataset configurations and 600 fresh generations. **Label validity:** the rewriting prompt that produces the most label flips is the one that instructs the model to change the label, so the probe is a different input rather than an attack. Reporting flip rate alone overstates measured attack success by $3.05\times$ overall and by $19.5\times$ on topic classification. **Input validity:** on sentence-pair tasks most rewrites collapse the two segments and are not well-formed classifier inputs; because malformed probes are scored as failures, leaving them in *understates* attack success, 12.5% against 41.7%. The two errors therefore push in opposite directions and do not cancel. **Measurement validity:** the reported metric must be a function of the generated text, a condition an earlier version of this study did not meet. Underneath these, LLM rewriting is a trade: it keeps only 44.5% of the attacks it is applied to while making them far more fluent (median perplexity 22.9 against 522). I close with three numbers any automated red-teaming report should carry.

1 Introduction

Automated red-teaming increasingly uses one language model to probe another. The attraction is obvious. Word-substitution attacks such as TextFooler (Jin et al., 2020) and PWWS (Ren et al., 2019) produce inputs that are effective but stilted, and a perturbation no fluent speaker would write says less about deployment risk than one that reads normally. An LLM can rewrite the whole sentence rather than swap tokens, and can therefore generate probes that are fluent by construction.

The generated probe, however, only tells us something about the target model if it is a valid test of that model. This paper asks how often it is, and what the answer does to the numbers such pipelines report.

A probe can fail to be a valid test in three distinct ways, and each one is invisible in the summary statistic that red-teaming reports usually carry:

- **Label validity.** The rewrite may have changed the input’s true label. Then a prediction change is correct behaviour, not a vulnerability.
- **Input validity.** The rewrite may not be a well-formed input for the task at all — for a sentence-pair task, not a pair.
- **Measurement validity.** The reported number may not be a function of the generated text, if the pipeline scores something else.

I measure all three. The first two are quantified on 600 generations produced for this paper under controlled prompt conditions; the third is demonstrated on a worked instance, an earlier version of this study, where it silently determined the published result.

Underneath the three failures is a simpler question that the same experiments answer. Rewriting is applied to attacks that *already succeed*, so its cost is directly measurable: how many survive being rewritten?

My contributions are:

- **Rewriting keeps under half the attacks it is applied to.** Across 32 configurations, LLaMA-2 rewrites of successful word-substitution attacks still flip the classifier 44.5% of the time, while being far more fluent (Section 3).
- **Label validity fails in the direction that flatters the attacker.** A task-specific prompt roughly doubles retention, but on topic classification it does so by instructing a topic change. Flip rate alone overstates attack success by $3.05\times$ pooled and $19.5\times$ on AG News (Section 4).
- **Input validity fails in the opposite direction.** Malformed sentence-pair probes are scored as failures, so not checking them understates attack success — 12.5% against 41.7%. An explicit structure constraint nearly doubles validity but does not fix it (Section 5).
- **I measure the generate-and-filter pipeline** proposed for this setting and report what each constraint removes; the similarity constraint, not fluency, does most of the work (Section 6).
- **A worked instance of measurement-validity failure**, and the one-line check that catches it (Section 7), followed by three numbers a red-teaming report should carry (Section 8).

2 Experimental Setup

Attacks and targets. I use TextFooler (Jin et al., 2020) and PWWS (Ren et al., 2019) as implemented in TextAttack (Morris et al., 2020), against `bert-base-uncased` and `roberta-base` classifiers fine-tuned on each dataset. Attacks were run over 500 test examples per configuration (277–500 after dataset exhaustion), yielding 190–479 successful attacks per configuration to sample from.

Datasets. Single-sentence tasks: IMDb, SST-2, MR (Rotten Tomatoes), AG News, and CoLA. Sentence-pair tasks: RTE, QNLI, and MRPC. Pair inputs are two segments joined by a `<SPLIT>` marker, following TextAttack’s convention.

Probe generation. Probes are produced by `Llama-2-7b-chat-hf` (Touvron et al., 2023) in half precision, with nucleus sampling ($p = 0.9$, $T = 0.8$, 160 new tokens), through the model’s chat template. I strip the scaffolding chat models place at both ends of an answer (“Here is the revised text:” and trailing explanations) before scoring; leaving it in would inflate perplexity and depress similarity for whichever prompt elicits more chattiness, which is the quantity under study.

Measurements. For each probe I record whether the target classifier’s prediction differs from its prediction on the original text (*flip*); cosine similarity between sentence embeddings of the original and the probe, using `all-MiniLM-L6-v2`; and perplexity under GPT-2-XL. For pair tasks I also record whether the probe preserved the two-segment structure and is a well-formed input at all.

The comparison of interest. Probes are generated only for examples on which the word-substitution attack already succeeded, so the baseline is fixed at 100% by construction and the quantity to report is what fraction of those attacks survives rewriting.

Dataset	Target	Specific	Generic	p
AG News	BERT	20/30	10/30	0.019
AG News	RoBERTa	19/30	8/30	0.009
IMDb	BERT	22/30	7/30	0.0002
IMDb	RoBERTa	28/30	6/30	$< 10^{-4}$
SST-2	BERT	12/30	13/30	1.000
SST-2	RoBERTa	18/30	14/30	0.438
Pooled		119/180	58/180	1.7×10^{-10}

Table 1: Probes that still flip the classifier, by prompt variant, with dataset and target held fixed. Two-sided Fisher exact test per cell.

3 Rewriting Keeps Under Half the Attacks

I first re-evaluate a stored corpus of rewrites spanning 32 configurations (2 target architectures $\times$ 2 attack recipes $\times$ 8 datasets), running each target classifier on the rewritten text.

On the 20 single-sentence configurations, rewrites still flip the classifier on average 44.5% of the time. Every one of these was a successful attack before rewriting, so more than half are destroyed by it. Retention varies enormously:

Dataset	Mean	Min	Max
AG News	0.907	0.706	1.000
IMDb	0.764	0.667	0.889
MR	0.268	0.077	0.462
SST-2	0.217	0.105	0.300
CoLA	0.069	0.000	0.133

The spread is a factor of thirteen, and it lines up with an implementation detail: AG News and IMDb were the two datasets for which a hand-written prompt template existed, while the others fell through to a generic one. Dataset and prompt are confounded here, so the next section separates them — and in doing so uncovers the first validity failure.

4 Label Validity

To break the confound I run *both* prompt variants on *every* dataset. The generic prompt is the original fallback (“Improve the text fluency while altering the classification. . .”). The task-specific prompts follow the shape of the hand-written ones: for IMDb and AG News these are the originals; for SST-2 I write one in the same shape, so that “has a task-specific prompt” becomes a manipulable variable rather than a property of the dataset. I sample 30 successful TextFooler attacks per cell.

Pooled over 360 probes, the task-specific prompt roughly doubles the fraction of attacks that survive: 119/180 against 58/180 ($p = 1.7 \times 10^{-10}$). The effect holds in 5 of 6 cells and is significant in 4; the two SST-2 cells show no reliable difference.

Why the prompt wins matters more than that it wins. An adversarial example must also preserve the input’s meaning, and therefore its true label. Splitting by task type shows two very different mechanisms behind the same doubling:

Dataset	Flip rate	Meaning-preserving	Ratio
AG News	65.0%	3.3%	19.5×
SST-2	50.0%	15.0%	3.3×
IMDb	83.3%	46.7%	1.8×
Pooled	66.1%	21.7%	3.05×

Table 2: Attack success under the task-specific prompt as a flip-rate-only report would state it, against the same probes required to keep similarity ≥ 0.75 .

	Flip rate	Similarity
<i>Sentiment (IMDb, SST-2)</i>		
specific	0.667	0.713
generic	0.333	0.705
<i>Topic (AG News)</i>		
specific	0.650	0.361
generic	0.300	0.691

On sentiment the specific prompt doubles the flip rate at no cost in similarity, 0.713 against 0.705. On AG News the same doubling arrives with similarity collapsing from 0.691 to 0.361 — and the reason is written in the prompt, which instructs the model to change the topic “from world to sports”. A text whose topic has actually changed is not an adversarial example. It is a different input with a different correct label, and a prediction change on it is the classifier behaving correctly.

How much this distorts a report. Table 2 puts a number on it. If a red-teaming pipeline reports flip rate alone, and we then ask how many of those flips came from probes that still mean what the original meant, the reported figure overstates measured attack success by 3.05× pooled — and by 19.5× on the task where the prompt instructs a semantic change.

This is a trap for the whole approach, and it is structural rather than accidental: **the instruction most effective at producing label flips is the instruction to change the label**. Any search over prompts that optimizes reported attack success will find it.

5 Input Validity

For RTE, QNLI and MRPC the classifier expects two segments. Most probes do not supply them: the model merges premise and hypothesis into a single passage, often appending a commentary sentence. In the stored 32-configuration corpus, only 100 of 198 pair-task rewrites preserved the structure; for the two RTE configurations under PWWS, *zero* of 31 did.

I test whether an explicit constraint fixes this, comparing the generic prompt against one stating that the input is a pair, that the segments are separated by `<SPLIT>`, and that the answer must return exactly two sentences in the same order.

Two things follow. First, stating the constraint nearly doubles structural validity, from 30.0% to 57.5%, and raises the yield of usable probes from 12.5% to 21.7% — but the model still violates an explicitly stated format more than 40% of the time, so the constraint is a mitigation and not a fix.

Second, and more consequential for anyone reading a red-teaming number: malformed probes are counted as attack failures, never as successes — none of the malformed probes registered a flip. Leaving them in the denominator therefore **understates** attack success, 12.5% against 41.7% for the generic prompt. Label-validity

Prompt	Well-formed	Unchecked	Valid only
generic	0.300	12.5%	41.7%
structured	0.575	21.7%	37.7%

Table 3: Sentence-pair tasks, 240 probes. *Well-formed*: fraction preserving the two-segment structure ($p = 2.8 \times 10^{-5}$ between prompts, Fisher exact). *Unchecked*: attack success if malformed probes are left in the denominator. *Valid only*: attack success among well-formed probes.

Stage	Kept	Frac.	Sim. / PPL
All probes	360	1.000	0.648 / 33.8
+ flips the classifier	177	0.492	0.596 / 28.1
+ similarity ≥ 0.75	64	0.178	0.840 / 27.2
+ perplexity ≤ 150	55	0.153	0.834 / 22.9

Table 4: Cumulative effect of the three conditions. Sim. is mean cosine similarity, PPL median GPT-2-XL perplexity, of the surviving set.

failure inflates the reported number and input-validity failure deflates it. **The two errors push in opposite directions, so they do not cancel**; which one dominates depends on the task, and neither is visible in the summary statistic.

6 What the Filter Trades

The standard response to unreliable generations is to filter them: keep only probes that flip the label, stay semantically close, and read fluently. I elaborate that filter into a per-candidate test — each probe scored on all three conditions and kept or dropped individually — and apply it to all 360 single-sentence probes.

The filter retains 15.3% of generations and does what it claims: mean similarity rises from 0.648 to 0.834. Two observations matter for practice. First, the *similarity* constraint does nearly all the filtering — it removes 113 of the 177 probes that flip, while the fluency bound removes 9 more. This inverts the usual framing, in which fluency is the reason to filter; here fluency is nearly free and meaning preservation is what costs.

Second, the fluency gain over the attacks being replaced is real and large. Over 428 attacked examples, the median GPT-2-XL perplexity of a TextFooler perturbation is 522, against 22.9 for filtered rewrites, and only 15.4% of 403 successful word-substitution attacks satisfy the same similarity and fluency constraints. **LLM rewriting buys a large improvement in fluency and pays for it in attack retention.** Neither side of the trade is small, and the exchange rate is set by the prompt.

7 Measurement Validity: A Worked Instance

The third failure is the one that most easily escapes notice, because it leaves no trace in the outputs. An earlier version of this study reported that filtered LLaMA-2 generations outperformed word-substitution attacks, with improvements up to 19.31% relative. Those numbers do not survive scrutiny, and the reasons are worth recording because the failure mode is easy to reproduce.

The classifier was never run on the probes. The pipeline wrote LLaMA output into a `llama_text` column, and the reported attack success rate was computed from TextAttack’s `result_type` field. That field describes the word-substitution attack, which ran *before* rewriting. The probes therefore never entered the reported numbers. Section 3 supplies the missing evaluation.

The headline table compared sample sizes. Each “filtered” figure was the attack’s success rate over 20 examples; each “baseline” figure was the same attack over 277–500. All five rows reproduce exactly from those two runs, so the reported improvement was a difference in sample size.

Filtering happened at a different layer than described. The pipeline did constrain and score generations, but not where the published algorithm places it. A similarity constraint, `min_cos_sim = 0.6`, was passed to the attack’s word-substitution step, and perplexity and grammar were computed afterwards as aggregate statistics over each result file. A per-candidate test on all three thresholds is not in that code, and the values 0.75 and 150 appear nowhere in it. Section 6 implements the per-candidate test.

The check that catches it. The lesson is narrow and mechanical: *if a pipeline produces new text, the reported metric must be a function of that text*. A one-line assertion — that the evaluated column is the generated column — would have caught all three problems. That this happened in a study whose entire subject was generated text is the point: measurement-validity failure is not a sign of carelessness about the topic, it is a default of the tooling, which will happily join a generations column to a metric computed from something else.

8 Three Numbers a Red-Teaming Report Should Carry

Each failure above is invisible in an attack success rate, and each is visible in one additional number:

1. **Semantic similarity beside every flip rate.** Without it, a prompt that changes the label scores best. On AG News this is the difference between reporting 65.0% and 3.3%.
2. **The fraction of generations that are well-formed inputs.** Without it, malformed probes sit in the denominator as failures. On sentence-pair tasks this is the difference between 12.5% and 41.7%.
3. **The number of generations actually scored by the target model.** Without it, a pipeline can report a metric computed from something other than what it generated, and nothing in the output will look wrong.

Limitations. Probes come from a single 7B model; larger or format-tuned models may preserve structure better. I sample 30 per cell, adequate for the effect sizes reported but not for small ones, which is why the SST-2 cells are inconclusive rather than negative. Similarity is measured with one sentence encoder, and encoder choice shifts how severe the 0.75 threshold is. The task-specific prompt for SST-2 was written by me in the shape of the originals; a different phrasing might move the effect. Finally, the three failures are demonstrated on classification; whether they take the same form in generative red-teaming is untested here.

9 Conclusion

An automated red-teaming result is a claim about a target model, but it is produced by a second model whose output may not be a valid test at all. I measured three ways that happens. A prompt can raise the flip rate by changing the input’s true label, which overstates attack success by $3.05\times$ pooled and $19.5\times$ on topic classification. A probe can fail to be a well-formed input, which understates it, 12.5% against 41.7%. And a pipeline can report a number computed from something other than what it generated, which is undetectable from the outputs alone.

Underneath, LLM rewriting is a trade rather than an improvement: it makes attacks dramatically more fluent, median perplexity 22.9 against 522, and destroys more than half of them. None of this makes the approach unusable. It makes it measurable, which requires reporting similarity beside flip rate, well-formedness beside both, and how many generations the target model actually saw.

Reproducibility

Code, the stored probes, and every result table in this paper are available at <https://github.com/yhj3/counterfactual-textattack>. The repository documents which components are original and which were reconstructed, and includes the audit described in Section 7.

References

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